Data and script files for the genome wide association studies for the 2015 salt experiments on the *indica* and *aus* Rice panels reported in Al-Tamimi et al. (2016)

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This document describes the data and code files that were used in the genome wide association study (GWAS) phase for the two experiments reported by Al-Tamimi et al. (2016): experiment 0254 that investigates an *indica* panel and experiment 0255 that investigates an *aus* panel. In addition, the combined single set panels (INDAUS) was subject to a GWAS. These files have been provided in GWAS_data_and_scripts.zip.

The Phenotypic data are the BLUEs produced in the phenotypic analysis phase by fitting Mixed Linear Models (MLM) to the phenotpic responses. This data and the scripts to produce it are in Phenotyping_data_and_script_files.zip. Genotypic data were composed of the HDRA dataset, a 700K SNP array designed by McCouchs laboratory at Cornell University (McCouch, 2016).

We used a GWAS approach referred to as the interaction model, amalgamates each phenotypic response for both treatments (salt and control). This approach uses fits an MLM that is fitted to includes the kinship (K) matrix as a random effect to control for population structure (Yang et al., 2014; Chen et al., 2011). When analyzing the combined population INDAUS, we also included principal components to control for population structure, because of the deep population structure in rice (Zhao et al., 2014; Yu et al., 2006).

1 Files in GWAS_data_and_scripts.zip

1.1 Files in the directory *Cleaned_SNP_and_kinship_data*

This folder contains the cleaned SNP data where only only SNP markers with a minor allele frequency of ≥ 0.05 and a number of accessions with a minor allele > 6 were used for association analyses. SNP data is in the Hapmap format, while kinship and principal component files were produced using TASSEL software (`http://www.maizegenetics.net/tassel`).

- Hapmap_AUS700.maf005.nomissing.hmp.txt – txt files containing the cleaned SNP data for aus panel.
- Hapmap_IND700.maf005.nomissing.hmp – txt files containing the cleaned SNP data for indica panel.
- Hapmap_INDAUS700.maf005.nomissing.hmp – txt files containing the cleaned SNP data for INDAUS panel.
- Kin_AUS700.maf005.nomissing – txt files containing the kinship data for aus panel.
- Kin_IND700.maf005.nomissing – txt files containing the kinship data for indica panel.
- Kin_INDAUS700.maf005.nomissing – txt files containing the kinship data for INDAUS panel.
- PC_INDAUS700 – txt files containing the principle component data.

1.2 Files in the directory *ASReml_interaction_R_scripts*

We developed a GWAS interaction model to account for the marker, and the interaction of the marker (SNP) and treatment (control and salt). The inclusion of the Marker-by-Treatment term, allows identifying markers that have a significant contribution depending on the treatment. The scripts contain code to be executed using the R statistical computing environment (R Development Core Team, 2015).

Data processing R scripts:

1. readindata_aus.R – an R script to import the aus txt files containing the raw data for cleaned SNP and kinship and amends files in the write format for the interaction model.
2. readindata_ind.R – an R script to import the indica txt files containing the raw data for cleaned SNP and kinship and amends files in the write format for the interaction model.
3. readindata_ausindica_pca.R – an R script to import the INDAUS txt files containing the raw data for cleaned SNP, kinship and principle component and amends files in the write format for the interaction model.

GWAS analysis R scripts: The R package `ASReml-R` (Butler et al., 2009) is used to perform the mixed model analysis for the GWAS, `qvalue` (Dabney and Warnes, 2004) is used to performs Bonferroni and false discovery rate (FDR) statistical tests and `qqman` (Turner, 2014) is used to produce the Manhattan plots.

1. analysedata_aus.R – an R script that uses the package `ASReml-R` to run the GWAS interaction model on the aus data without principle component analysis (PCA).
2. analysedata_ind.R – an R script that uses the R package `ASReml-R` to run the GWAS interaction model on the indica data without principle component analysis (PCA).
3. analysedata_ausind. R – an R script that uses the package `ASReml-R` to run the GWAS interaction model on the INDAUS data with principle component analysis (PCA).
4. map.R — an R script that adds the chromosome and base pairs columns into the output of analysedata.R in the right order.
5. Manplots.R — an R script that uses the R package `qvalue` to perform Bonferroni and false discovery rate (FDR) statistical tests and uses the R package `qqman` to plot the Manhattan plots for all traits.

1.3 Files in the directory *Data_files_produced_from_R_scripts*

The files in this folder are produced as a result of executing the R scripts in the directory *ASReml_interaction_R_scripts*.

1. Output_readindata_aus – data files produced from executing the script readindata_aus.R.
2. Output_readindata_ind – data files produced from executing the script readindata_ind.R.
3. Output_readindata_ausindica_pca – data files produced from executing the script readindata_ausindica_pca.R.
4. Output_analysedata_aus – data files produced from executing the script analysedata_aus.R.
5. Output_analysedata_ind – data files produced from executing the script analysedata_ind.R.
6. Output_analysedata_ausind_pca – data files produced from executing the script analysedata_ausind.R.
7. Output_Maps_aus, Output_Maps_indica and Output_Maps_INDAUS – data files produced from executing the script map.R.
8. Output_Manplots_aus, Output_Manplots_indica and Output_Manplots_INDAUS – data files produced from executing the script Manplots.R.

References

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